[Total: 6]

**3** In a system called the positronium, an electron and positron move in a circular orbit as shown in Fig. 3.1. A positron has a positive elementary charge *e* and a mass identical to that of an electron. The positron and electron are separated by a distance of $1.30 \times 10^{-10}$ m.

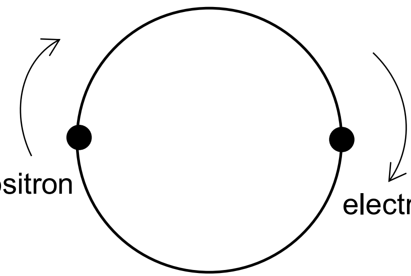


**Fig. 3.1**

**(a)** Explain why the centripetal forces on the electron and positron are equal in magnitude.

The force that exist between the 2 charges is the electric force and by newton's 3'rd law, they have the same magnitude. Since the only force that is exerted on the positron or electron is the electric force and this provides the centripetal force, the 2 centripetal forces have the same magnitude. (Gravitational force between the 2 masses is very small and thus negligible.) [1]

**(b)** Determine

**(i)** the magnitude of each centripetal force,

The electric force provides the centripetal force,

$$F = \frac{q_1 q_2}{4\pi\varepsilon r^2}$$
$$= \frac{(1.60x10^{-19})(1.60x10^{-19})}{4\pi(8.85x10^{-12})(1.30x10^{-10})^2}$$
$$= 1.36x10^{-8} N$$

centripetal force = .......................................... N [2]

**(ii)** the time taken for one revolution of the electron.

$$F = mr\omega^2$$
$$1.36x10^{-8} = (9.11x10^{-31})(\frac{1.30x10^{-10}}{2})(\frac{2\pi}{T})^2$$
$$T = 4.15x10^{-16} s$$

time taken = .......................................... s [2]

**(c)** State and explain whether there would be any difference to the circular orbit if it were a proton in place of the positron.

When the positron is replaced by a proton, the electric force remained unchanged (centripetal force is also unchanged) but the mass of the proton is much higher (2000 times) than the electron, thus, the radius of the orbit for the proton will be much smaller than that of the electron. The result will be the proton seems to be stationary while the electron will orbit around the proton. [2]

[Total: 7]